

GENERAL SPECIFICATION G4.5

Compressed Air Systems

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G4.5 COMPRESSED AIR SYSTEMS

G4.5.1 General Requirements

- (a) Compressed air supply plant shall be designed and configured according to the application(s) which are classified into the following groups:
 - (i) A: Motive power source for operating valve actuators, air motors, reverse jet filters, silo arch breaking pads, pneumatic massagers and pneumatically operated machines;
 - (ii) B: Motive power source for starting diesel generating sets;
 - (iii) C: Instrument air supply;
 - (iv) D: Air scour of membrane filters (if specified); and
 - (v) E: Air feed to operations such as dissolved air flotation, biological aeration and pumping surge suppression (if specified).
- (b) Unless otherwise specified and except for group (B) applications, compressed air supply plant shall include at least one duty and one standby compressor set which shall be electrically driven by A.C. squirrel-cage induction motors.
- (c) Air shall be filtered to remove dust to protect the compressors from undue wear and to suit the needs of the application. Filtration shall take place on the intake side of the compressors and in the case of group (C), additionally also at point of use of the air.
- (d) Oil lubricated compressors may be used for group (A), (B) and (C) applications if oil eliminating filters are installed on the delivery side of the compressors.
- (e) Where oil mist lubrication is recommended for certain types of valve actuator, the lubricators shall be installed at each point of use of the air. Oil-free compressors shall be used for group (D) and (E) applications.
- (f) Air shall be cooled by means of after-coolers except generally in the case of group (D) and (E) applications.
- (g) Air receivers shall be provided for group (A), (B), (C) and (E) applications to balance supply and demand or to provide a reserve to operate a prime mover.
- (h) Air dryers shall be provided for group (A) and (C) applications and when recommended for group (D) to suit the membrane filters.
- (i) For the avoidance of doubt, Normal Air Conditions shall be defined as follows:
 - (i) Pressure: 1 atm.
 - (ii) Temperature: 20 deg. C. (293.15 deg. K.)
 - (iii) Relative humidity: 0 %and the units of air flowrates shall use the capitalised alphabet 'N' such as in 'Nm³/hr', etc. to denote Normal Air Conditions. The above shall be for both air and gas applications.
- (j) Free Air Delivery (FAD) shall be defined as the volume of compressed air leaving the compressor discharge and expanded back to 1 bar (atm), 20 deg. C. and 0 % relative humidity.

- (k) Compressor efficiency values shall be accompanied by the tolerance limits on the output flowrate, pressure and temperature with respect to the power consumption as a minimum.
- (l) Noise from compressed air plant shall be limited as set out in the General Specification Section of Workmanship Requirements and as per the WSH Act.
- (m) Parts of the compressed air supply plant other than the compressor sets and drivers where the surface temperature routinely exceeds 60°C shall be insulated and cladded with aluminium.
- (n) Water-cooled components subject to condensation shall also be insulated and cladded in the same way.
- (o) Requirements for blower applications shall refer to the General Specification on Blowers.

G4.5.2 Air Intake

- (a) Air intakes shall be positioned in a manner which avoids collection of contaminants such as engine exhaust fumes and excessive dust and to be always away from any extractor fan discharge points.
- (b) Air may be admitted to the room which houses the compressors either through an external wall or, where this is not possible, via an intake duct. Where air is delivered to processes on a continuous basis, the compressors may be connected to the intake duct to draw air from outside the building and not from the room itself. The Contractor shall submit duct sizing calculations taking account of all losses to justify duct dimensions selection and based on the duct material.
- (c) A louvered panel or panels shall be installed at the air inlet point together with filters. The filters shall consist of removable panels with convoluted elements mounted in frames. The elements shall be of the washable type and a spare set shall be provided to facilitate substitution of the filters being washed.
- (d) Intake ducting shall be in galvanised steel and of adequate proportions to minimise pressure drop and vibration. Where necessary to suppress low frequency noise, a silencer of the multiple aerofoil or 'beam splitter' type shall be installed within the duct close to the inlet point.
- (e) Where air is drawn by the compressors directly from the room, appropriate provision shall be made to protect equipment and pipework from low ambient temperatures. Where air is drawn from outside the building, heat from the machines must be dissipated effectively at times of high ambient temperatures.
- (f) Where applicable, the Contractor shall implement means of drawing out heated air via ductwork from the compressors' surroundings to the outside of buildings to prevent heat build-up.
- (g) The room shall be properly ventilated.

G4.5.3 Reciprocating Compressors

- (a) Reciprocating compressors of the oil-lubricated type shall be employed only for application groups (A), (B) and (C) as defined in Section G4.5.1.

- (b) Oil-free reciprocating compressors may be employed for other low capacity applications.
- (c) Unless otherwise specified, each compressor shall be rated for continuous operation at twice the maximum air demand or to recharge the receiver from empty in one hour whichever is greater.
- (d) They shall be suitable for intermittent operation and shall maintain the associated air receiver pressure between pre-set limits irrespective of actual air demand up to the maximum demand.
- (e) Cylinder heads shall be readily removable for inspection.
- (f) Each compressor shall be driven via a Vee-belt drive arrangement by a cage induction motor, which shall be rated to allow the compressor to operate at the safety valve setting without overloading.
- (g) The compressor and motor shall be mounted on a combination baseplate with slide rails for belt tension adjustment. A sheet steel guard shall be bolted to the baseplate.
- (h) Where necessary, each compressor shall have an automatic solenoid operated unloading system to facilitate starting.
- (i) This system shall be initiated by auxiliary contacts on the compressor motor starter and shall be subject to an adjustable time delay.
- (j) Oil-lubricated machines shall have an automatic oil lubricator fitted with indication of oil level and feed rate. The delivery side of each machine shall have an oil-eliminating filter with provision for draining into a portable container which shall be provided.
- (k) Each compressor shall have an inlet air filter and silencer, a pressure safety valve, fusible plug, check valve and isolating valve.
- (l) Compressors shall be tested in accordance with BS ISO 1217 and BS 1571 Part 2.
- (m) Safety valves shall comply with BS EN ISO 4126-1. The installation of safety valves, gauges and fusible plugs shall comply with BS 1123.

G4.5.4 Rotary Compressors

- (a) Rotary compressors shall be selected from the following types unless otherwise specified:
 - (i) Rotary lobe (Roots) compressors;
 - (ii) Liquid ring compressors; and
 - (iii) Screw compressors.
- (b) Rotary Lobe Compressors
 - (i) Generally, rotary lobe machines shall be sized to deliver directly the necessary flow rate of air at the pressure required by the media filter or by the process. Due allowance shall be made for operating a pressure safety valve set at a higher pressure without overloading the compressor or its driver.

- (ii) Under no circumstances shall a compressor be selected to operate at or above the continuous maximum temperature or discharge pressure recommended by the compressor manufacturer.
 - (iii) A clear safety margin of at least 0.2 bar shall be allowed.
 - (iv) The sealing arrangement between the timing gearbox and compression chamber shall ensure that leakage of oil cannot take place during operation or shutdown.
 - (v) Where necessary to facilitate starting of the compressor (e.g. at reduced voltage), an unloading arrangement shall be incorporated in the compressor delivery system and shall include a motorised valve operated via auxiliary contacts on the compressor motor starter.
 - (vi) Each compressor shall be driven via a Vee-belt or direct drive arrangement by a cage induction motor.
 - (vii) The motor may be arranged for variable speed or two speed operation where this suits the requirements of the application. The compressor and motor shall be mounted on a combination baseplate with slide rails for belt tension adjustment.
 - (viii) A sheet steel guard shall be bolted to the baseplate.
 - (ix) Each compressor shall have an inlet air filter, inlet silencer, discharge silencer, pressure safety valve, pressure gauge, check valve and isolating valve. In process applications where the discharge pressure may approach the maximum permitted, the machine shall be arranged to shut down on high discharge temperature and a suitable switch and thermometer shall be provided.
 - (x) Where appropriate, the compressor and its accessories may be pre-assembled on a steel frame.
 - (xi) Compressors shall be tested in accordance with BS ISO 1217 and BS 1571 Part 2.
 - (xii) Safety valves shall comply with BS EN ISO 4126-1. The installation of safety valves, gauges and fusible plugs shall comply with BS 1123 Part 1.
- (c) Liquid Ring Compressors
- (i) Liquid ring compressors may be used for, in special circumstances only, for group (D) as defined in Section G4.5.1 and where specified in the Particular Specification.
 - (ii) Generally, liquid ring machines for process use shall be sized to deliver directly the flow rate of air at the pressure required by the process. Due allowance shall be made for operating a pressure safety valve set at a higher pressure without overloading the compressor or its driver.
 - (iii) Each compressor shall have a seal water supply system, which shall include an isolating valve, in-line strainer, solenoid-operated valve, flow switches, regulating valve and check valve. The solenoid-operated valve shall be operated by auxiliary contacts on the compressor motor starter. The flow switches shall initiate a shutdown of the compressor if the seal water flow goes outside high or low limits following start-up.

- (iv) The compressor materials shall be compatible with the seal water quality. Generally, a cast iron casing and grade 316 stainless steel rotor and shaft shall be used unless otherwise specified. Bearings shall be of the outboard type and shaft sealing shall be effected by packed glands.
 - (v) Each compressor shall be driven via a V-belt, mechanical coupling, magnetic or alternative drive arrangement by a A.C. squirrel-cage induction motor normally arranged for fixed speed operation.
 - (vi) The compressor and motor shall be mounted on a combination baseplate with slide rails for belt tension adjustment. A sheet steel guard shall be bolted to the baseplate.
 - (vii) The delivery side of each compressor shall be connected to a dedicated water separation vessel. The vessel shall be of welded carbon steel construction to PD 5500 and equipped with supporting feet, access manway, sight glass with safety enclosure, float-operated drain valve, manual drain valve, pressure gauge and pressure safety valve.
 - (viii) The vessel shall be designed to maximise dis-entrainment of water and where necessary shall be fitted with suitable baffles.
 - (ix) Level switches shall be provided to initiate shutdown of the compressor if the water level in the separation vessel goes outside high or low limits.
 - (x) The used seal water shall be piped to an approved discharge point.
 - (xi) Each compressor shall have an inlet silencer, inlet check valve, discharge check valve and isolating valve in addition to the features detailed above.
 - (xii) Compressors shall be tested in accordance with BS ISO 1217 and BS 1571 Part 2.
 - (xiii) Safety valves shall comply with BS EN ISO 4126-1. The installation of safety valves, gauges and fusible plugs shall comply with BS 1123.
- (d) Screw compressors
- (i) Screw compressors of the oil-free type may be used for application group (E) as defined in Section G4.5.1.
 - (ii) Generally, screw compressors for intermittent duty shall be rated for continuous operation at twice the maximum air demand and shall be suitable for intermittent operation to maintain the associated air receiver pressure between pre-set limits irrespective of air demand up to the maximum value.
 - (iii) Screw compressors for continuous duty shall be sized to deliver directly the flow rate of air at the pressure required by the process. Due allowance shall be made for operating a pressure safety valve set at a higher pressure without overloading the compressor or its driver.
 - (iv) Under no circumstances shall a compressor be selected to operate at or above the continuous maximum temperature or discharge pressure recommended by the compressor manufacturer. A clear safety margin shall be allowed.
 - (v) The sealing arrangement between the timing gearbox and compression chamber shall ensure that leakage of oil cannot take place during operation or shutdown.

- (vi) Each compressor shall be driven via a V-belt drive or alternative mechanical coupling arrangement by a A.C. squirrel-cage induction motor.
- (vii) The motor may be arranged for variable speed operation where this suit the requirements of the application. The compressor and motor shall be mounted on a combination baseplate with slide rails for belt tension adjustment. The skid shall be mounted on a plinth. A sheet steel guard shall be bolted to the baseplate.
- (viii) Each compressor shall have an inlet air filter, inlet silencer, discharge silencer, pressure safety valve, pressure gauge, check valve and isolating valve. The machine shall be arranged to shut down on high discharge temperature and a suitable switch and thermometer shall be provided.
- (ix) Compressors shall be tested in accordance with BS ISO 1217 and BS 1571 Part 2.
- (x) Safety valves shall comply with BS EN ISO 4126-1. The installation of safety valves, gauges and fusible plugs shall comply with BS 1123.

G4.5.5 Centrifugal Compressors

- (a) Centrifugal compressors shall be the single stage centrifugal type driven through speed increasing gear unit by an electric motor and shall be factory mounted on a common base plate with all accessories required for a complete system. Each system shall include a discharge check valve.
- (b) Single stage compressors shall be horizontally split, single stage units with overhung shaft design or impellers mounted between bearings, integral suction and tangential discharge. Motor shall be generally not exceed 1500 rpm totally enclosed fan cooled (TEFC) squirrel cage induction type.
- (c) Each compressor shall develop a discharge pressure at least 0.05 bar above the specified discharge pressure when the capacity is reduced to 70% of the capacity at the specified discharge pressure.
- (d) Compressors shall operate satisfactory without surging at any point between 60 and 100% of the specified capacity.
- (e) Casings shall be cast iron made in sections with joints between sections accurately machined. Sections shall be held together by means of steel tie rods.
- (f) Seals shall be provided where the shaft passes through the ends of the casing. Suitable seals shall also be provided to prevent bearing contamination and to prevent loss of bearing lubricant.
- (g) Suction and discharge nozzles shall be flanged. Flanges shall be flat faced with diameter and drilling to BS EN 1092.
- (h) Each section of the casing shall be equipped with a casing drain. Casing drains shall be piped together, and the common drain line piped to the edge of the baseplate. A drain valve shall be provided.
- (i) Impellers may be of either fabricated or cast construction. Fabricated impellers shall be made of aluminium or grade 316 stainless steel and shall be riveted to cast hubs.

- (j) Cast impellers shall be accurately formed and machined on all exterior surfaces. Each impeller shall have uniform sections, smooth surfaces, be free from cracks and porosity and shall be dynamically balanced.
- (k) Each shaft shall be of solid design, fully machined or ground to size and polished. Each shaft shall be of ample size to transmit the maximum applied power and to carry all applied radial loads without excessive deflection.
- (l) Each rotor assembly shall be supported at each end by one or more oil or grease lubricated anti-friction bearings.
- (m) All points where oil leakage may occur shall be suitably trapped to prevent oil contamination of air. There shall be a built-in oil leak collection tray below the compressor unit for oil containment.
- (n) Bearings shall have a L-10 life rating of 100, 000 hours.
- (o) Shaft speed shall not exceed the bearing manufacturer's recommendations. Bearings shall be mounted in housings outboard of the casing heads.
- (p) Bearing housings shall be arranged to permit inspection and replacement of the bearings without disconnecting the piping or disassembling the compressor.
- (q) Each compressor shall be directly connected to the drive motor by means of a flexible coupling. Each coupling shall have a power rating of not less than 1.25 times the motor nameplate power when the misalignment is within the coupling manufacturer's tolerance limit. A suitable coupling guard shall be provided.
- (r) Each compressor shall be equipped with a dry type air filter located upstream of the suction inlet and silencers both in the suction and delivery.

G4.5.6 Aftercoolers

- (a) All compressed air supply plant other than group (D) applications and where liquid ring compressors are used shall incorporate after-cooling.
- (b) Unless otherwise specified, a single aftercooler with isolating and bypass arrangement shall be provided for applications (A), (B) and (C) only as defined in Section G4.5.1. In other applications, standby aftercooler is required.
- (c) Aftercoolers shall either be of the air-cooled type with finned-tubes and electrically driven cooling fan or the water-cooled shell and tube heat exchanger type.
- (d) Materials in contact with cooling water shall be resistant to corrosion and chemical cleaning fluids. All heat exchange surfaces shall be accessible for cleaning.
- (e) Aftercoolers of the shell and tube heat exchanger type with air on the tube side and water on the shell shall be designed to BS 853-2 and shall be type 2.
- (f) The heat exchanger shell shall be in carbon steel 151 360 A to BS EN 10028 Parts 1 to 3 and BS EN 10029 and the corrosion allowance shall be not less than 3 mm.
- (g) The shell shall be blast cleaned and coated internally and externally as set out in the General Specification on Painting and Protective Coatings. The epoxy system shall be used for internal parts of the shell and for external surfaces also. The heat exchanger tubes shall be copper designation C106 to BS EN 12451 or, where specified, 70/30 cupronickel designation CN107 to BS EN 12451.

- (h) The shell and tube side classifications shall be not less than Class 150.
- (i) Pipe connection flanges shall be drilled to BS EN 1092 with the appropriate pressure designations but not less than PN10. Provision shall be made for removal of condensate.
- (j) The heat exchanger shall be thermally insulated and cladded with aluminium.
- (k) The air outlet pipework shall be fitted with a thermometer and a temperature switch with voltage free changeover contacts to initiate a high temperature alarm. The water supply pipework shall be fitted with a flow meter and there shall be a pressure gauge installed in the piping prior inlet connection to the heat exchanger.
- (l) Generally, the air outlet temperature shall approach the cooling medium temperature to within 10 deg. C at maximum air throughput or as otherwise specified.

G4.5.7 Intercoolers

- (a) Requirements for intercoolers shall be similar as for aftercoolers.

G4.5.8 Condensate Separators

- (a) Condensate separators shall be installed downstream of air aftercoolers.
- (b) The separators shall consist of pressure vessels designed to PD 5500 and fabricated from carbon steel 151 360 A to BS EN 10028 Parts 1 to 3 and BS EN 10029. The internal trim shall be designed to remove moisture from the air in droplet and mist form.
- (c) The separator shall be equipped with an automatic drain valve with isolating valve and separate manual drain valve and voltage-free contact for high level alarm. The drain connections shall discharge to a designated location approved by the S.O..
- (d) The air outlet from the separator shall be fitted with a thermometer and temperature switches with voltage free changeover contacts to initiate high and low temperature alarms.

G4.5.9 Air Receivers

- (a) Air receivers shall be installed wherever it is necessary to balance a variable rate of air consumption with a fixed rate of supply which includes applications in groups (A), (C), (E) or where an air reserve is needed to operate a prime mover such as in group (B) and sometimes in group (A) as defined in a preceding section.
- (b) The capacity shall be sufficient to limit the number of compressor starts to no more than 10 per hour or as recommended by the manufacturer, whichever is lesser. In group (B) applications, the capacity shall be sufficient for the specified number and duration of engine start attempts.
- (c) Air receivers shall be of welded steel construction complying to BS EN 286-1 and PD 5500 with a corrosion allowance of 2 mm and may be either vertically or horizontally mounted according to capacity. The vessels shall have suitable supporting feet or cradles as appropriate.
- (d) The air receivers shall be painted according to the applicable painting system specified in the General Specification – Painting and Protective Coatings.

- (e) Air receivers supplied under this contract shall comply with the requirements of local regulating authorities.
- (f) The fittings shall include a pressure gauge, pressure safety valve, automatic drain valve with isolating valve and separate manual drain valve. The drain connections shall discharge to a designated location approved by the S.O..
- (g) Safety valves shall comply with BS EN ISO 4126-1. The installation of safety valves, gauges and fusible plugs shall comply with BS 1123.
- (h) Pressure switches with voltage-free contacts shall be installed on the receiver to control the associated compressors and to initiate a low-pressure alarm. The control relays shall normally be mounted in the same panel as the compressor motor starters. Both the cut-in and cut-out pressure settings of the standby machine(s) shall be set lower than the corresponding settings of the duty machine(s).
- (i) The air receivers shall be registered with the Ministry of Manpower and the registration number shall be painted on the air receiver itself. The Contractor shall be responsible for conducting all the required tests and meeting other requirements to achieve this.

G4.5.10 Air Dryers

- (a) Air dryers shall be of the automatic twin cell desiccant type designed to produce an outlet dew point better than -40 deg. C for applications in groups (A) and (C) and, when required, for group (B) as defined in Section G4.5.1. Dew point indicator-transmitter panel shall be located near to the dryers.
- (b) The dryer shall be of the floor mounting or wall-mounting type with twin carbon steel adsorption vessels packed with silica gel beads. Dryer vessel shall have provision of opening(s) of dimensions sized adequately to facilitate removal of hardened and clumped desiccant.
- (c) Dryers shall be designed and tested to BS ISO 7183 as applicable.
- (d) The vessels shall each have an electric immersion heater designed for a 230 V AC supply unless otherwise specified, air inlet and outlet connections and drying media filling and emptying platforms. All technical specification details of the desiccant shall be furnished to facilitate future procurement and replacement purposes.
- (e) The dryer shall incorporate an automatic adsorption changeover and reactivation system consisting of solenoid-operated valves and a control panel. The air for reactivation purposes shall be taken from the dry air outlet side and passed through the adsorber to be reactivated at a rate controlled by an orifice plate.
- (f) Air dryer adsorber units that are on duty shall be sized for 8 hours drying capacity when operating at pressures between 4.5 and 7 Bar (abs). Duration for regeneration shall not exceed 8 hours.
- (g) The dryer shall be thermally insulated and clad with aluminium sheet.
- (h) The reactivation cycle shall include a period of air purge with the heater deactivated to cool the reactivated adsorber to the operating temperature prior to changeover.

- (i) The control panel shall incorporate the following features:
 - (i) Isolating switch for 230 V AC supply;
 - (ii) Circuit protection for 230 V AC and 24 V DC circuits;
 - (iii) Cycle timer;
 - (iv) Heating timer;
 - (v) Heater contactors;
 - (vi) Control relays;
 - (vii) Control transformer;
 - (viii) LED Indicator lamps for CELL No 1 DUTY, CELL No 2 DUTY, HEATER No 1 ON, HEATER No 2 ON;
 - (ix) LED Indicator lamp for DRYER FAULT, voltage-free contacts for remote transmission of same shall be connected to the SCADA system;
 - (x) ON/OFF push-button(s);
 - (xi) RESET push-button; and
 - (xii) LAMP TEST push-button.

G4.5.11 Particle Filters

- (a) Particle filters on process gas streams shall be designed to remove dust with the efficiency of greater than 99.9% at particle size of 1.0 micron, 99.5% at a particle size of 0.5 micron and 98% at a particle size of 0.1 micron.
- (b) Any other air quality requirement with respect to particles, water and oil content in the compressed air shall be to fulfil the specific need of the process in question and to the air quality classes defined in BS ISO 8573-1.
- (c) The pressure drop through the filter at the maximum gas flow rate shall not be greater than 0.03 bar when the filter is clean and not greater than 0.1 bar when the filter is dirty.
- (d) The filters shall consist of carbon steel bodies with flanged inlet and outlet connections and removable top cover providing access to replaceable cartridge elements.
- (e) Sufficient elements shall be installed so that replacement will be needed not more frequently than once every six months.
- (f) Where specified, a duplex arrangement of filters with inlet and outlet isolating valves shall be provided to permit element replacement without interruption to the operating plant.
- (g) The filters shall be equipped with a differential pressure indicator and switch with voltage free changeover contacts, which shall be connected to the SCADA system to initiate a high pressure drop alarm.
- (h) A manual sampling and purge valve shall be provided downstream of the filter to permit testing with or without gas feed to the downstream process.

G4.5.12 Air Pipework

- (a) Air pipework includes piping for Air High Pressure (AHP), Air Low Pressure (ALP), Aeration Air (AA), Scour Air (AS), Instrument Air (IA) and Foul (Odourous) Air (FA). The materials of construction for the pipework of these applications are as follows:
 - (i) AHP: SS316 with pipe wall thickness schedule based on pressure rating of at least 10 Bar;
 - (ii) ALP: SS316 with pipe wall thickness schedule based on pressure rating of not more than 10 Bar;
 - (iii) AA: SS316 with pipe wall thickness schedule based on pressure rating according to process requirements for non-submerged applications;
 - (iv) AA: CPVC with pipe wall thickness schedule based on pressure rating according to process requirements for submerged applications;
 - (v) AS: SS316 with pipe wall thickness schedule based on pressure rating according to process requirements;
 - (vi) IA: SS316 with pipe wall thickness schedule based on pressure rating according to process requirements; and
 - (vii) FA: FRP with pipe wall thickness schedule based on pressure rating according to process requirements for submerged applications.
- (b) The pressure ratings for the above piping materials shall be with the allowance factors as specified in the General Specification Pipework.
- (c) Detail specifications of the above pipe materials are elaborated in General Specification G4.2 Pipework.
- (d) Pipework shall be arranged to minimise the transmission of vibration and where necessary, connections shall be made with anti-vibration couplings. Additional pipes support and fittings shall be mounted to the wall as required and approved by S.O..
- (e) Care shall be taken at points where condensation is likely to form or to accumulate such as following compression or cooling of ambient air or discharge of reactivation air from dryers.
- (f) All main pipes and branch pipelines shall slope downward at a gradient of at least 1/1200 in the direction of air flow to allow condensation to flow down drip legs and collect at the low points for removal. An automatic trap or drain valve shall be installed at the bottom of the drip leg to discharge the condensate to a designated location approved by the S.O.. Branch pipelines tapping points shall be from the crown instead of invert of main pipes. Unless otherwise specified, all pipe joints shall be welded to minimise leakage.
- (g) Thermal insulation and cladding where fitted shall be arranged in a manner which provides access to valves and other components for maintenance purposes.

G4.5.13 Acoustic Enclosure

- (a) The enclosure shall be selected based on its noise reduction performance, maintainability and its ability to withstand the elements.
- (b) Unless otherwise specified, the noise attenuation/reduction shall achieve a sound level of below 85 dBA, measured at 1 m around the exterior of the silencer.

- (c) To facilitate major overhaul of equipment or components, the enclosure shall be designed to allow for partial disassembly to allow access to the compressor and motor by a lifting device. The enclosure shall be fitted with lifting points for complete removal.
- (d) The enclosure shall incorporate over temperature detection device the output of which shall be suitable for control and monitoring.
- (e) The enclosure shall be provided with a suitable method of ventilation for compressor cooling. If ventilation fans are required, duty / standby ventilation fans shall be provided with proper ductwork inside the acoustic enclosure to direct all exhaust air to outside of the building. The fans shall be temperature controlled with temperature sensors designed to signal an alarm in the event of the ventilation fan failure due to over temperature.
- (f) The enclosure shall incorporate hinged doors or lift-off panels to provide safe access to the air compressor system components requiring inspection, cleaning or maintenance. The structural integrity of the enclosure shall be maintained when one or more panels are removed. Doors/panels shall comply with the following requirements as a minimum:
 - (i) Doors shall open to a minimum angle of 90° and be restrained against wind loads when open by fitting stay put type stays;
 - (ii) Safety warning signs such as 'Warning – Rotating Machinery Within', etc. shall be put up at a conspicuous position on the enclosure door;
 - (iii) Panels shall be easily removed and manipulated by one person (e.g. fitted with suitable handles) and have a maximum weight of 25 kg. Any lift-off panel weighing in excess of 25 kg owing to special design requirements or located at difficult to access locations owing to space constraints shall be clearly identified and lifting device such as mobile lifting A-frame or monorail hoist shall be provided to facilitate ease of the panel removal;
 - (iv) All keys and special tools for opening and lifting of enclosure panels shall be provided;
 - (v) Doors/panels shall be securely held in place whether in the closed or opened position and shall be lockable by padlock; and
 - (vi) Seals shall be provided around doors/panels to maintain the acoustic performance of the enclosure.
- (g) Where 'walk-in' type acoustic enclosures are proposed, these shall be provided with all appropriate warning signs/notices and sufficient small power and lightings, subject to the S.O.'s approval.
- (h) All routine inspection and servicing shall be achievable without the need to remove the enclosure.
- (i) Enclosure discharge ventilation and pressure relief valve discharge (if applicable) shall be ducted or vented away from the enclosure so that the high temperature air is dissipated away from the enclosure and any personnel access.
- (j) All discharge air duct and discharge air shall be acoustically treated and ducted outside the building to reduce the noise level as specified and meeting the latest regulations.

G4.5.14 Air Isolating Valves

- (a) Air isolating valves shall be generally of the cast iron gate type to BS EN 1171 or butterfly type to BS EN 593 and complying with the requirements of the General Specification on Valves, Penstocks and Stoplogs.
- (b) Air isolating valves up to 50 mm nominal bore diameter may alternatively be in copper alloy to BS EN 12288 where compatible with the required temperature and pressure rating.
- (c) The Contractor shall make a thorough assessment of the environment of concern to evaluate whether stainless steel SS316L grade isolating valves may be required with respect to any tendency for copper material to tarnish or corrode. The assessment results shall be submitted to the S.O. for approval and where necessary, SS316L valves shall be used instead of copper alloy isolating valves.
- (d) All isolating valves shall be tested in accordance with BS EN 12266.
- (e) Electric actuators for valves and solenoid-operated valves shall comply with the requirements of General Specification on Valves, Penstocks and Stoplogs.

G4.5.15 Air Flow Control Valves

- (a) Air flow regulating valves shall be globe type constructed of stainless steel 316C16 to BS EN 10213. Valve spindle shall be stainless steel type 316S16 to PD 970. Gland packing and seating ring shall be PTFE. Valves shall be sized in accordance with BS EN 60534 – 2. Flanged valves shall have integral flanges drilled to BS EN 1092 with pressure designation to match the pipework system. The flow control range of the valves shall be 10:1 or greater.

G4.5.16 Air Filters and Silencers

- (a) Air filters on the inlet to compressors shall be of the dry element two stage, high volume, and pleated type and shall have differential pressure gauge to determine when cleaning or element replacement is required. Each air filter shall be with automatic condensate drain and the condensate outlet nozzle shall be piped to a designated location approved by the S.O..
- (b) The intake silencer shall be located downstream of the filter and it shall be acoustical absorption type with fibre glass packing.
- (c) Unless otherwise specified, the noise attenuation/reduction shall achieve a sound level of below 85 dBA, measured at 1 m around the exterior of the silencer.

G4.5.17 Control of Instrument Air Compressors

- (a) The control system shall be designed so that either compressor may be selected as the duty unit and the other compressor as the stand-by unit. This selection shall be made locally and indicated on the SCADA system.
- (b) The compressor shall be controlled automatically by the pressure-unloading valve associated with it to maintain the pressure in the air receivers within the limits necessary for the satisfactory operation of all associated plant.

- (c) Compressors shall be fully controlled from a local panel with only monitoring provided on the SCADA system unless otherwise specified in the Particular Specification. An available signal shall be provided for each air compressor and these shall be connected to the SCADA system for indication and alarming.
- (d) The stand-by compressor shall start automatically and deliver air to the air receivers if the pressure in the air receiver falls to a preset value, which shall be above the minimum value necessary for the satisfactory operation of all associated plant. The stand-by compressor delivery shall be controlled automatically by its own pressure unloading valve.

END OF SECTION